ATHARVA AUTI

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EDUCATION

University of Southern California (M.S. in Cybersecurity Engineering) Grade A, 3.67/4.0 *Aug 2024 – Aug 2026*

- **Relevant Coursework: INF 519** Foundations & Policy for Information Security, **CSCI 530** Security Systems, **INF 523** Computer Systems Assurance, **ITP 475** Advanced Digital Forensics, **INF 529** Security & Privacy, **INF 517** Research Methods for User Studies

Mumbai University (Bachelor of Engineering in Cybersecurity, Minors – AI ML) Grade A, 9.51/10 *Aug 2020 - Jun 2024*

- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, OOP Systems, Computer Architecture, Cryptography, Database Management & Security, Web Application Security, Network Security, Virtualization & Cloud Computing, Cyber Laws

EXPERIENCE

Ernst & Young (EY) U.S. LLP *(Cybersecurity Consultant Intern)*, Los Angeles, CA *Jun 2025 – Aug 2025*

- Contributed to a **Fortune 100** retail client's global **Vulnerability Management** program, adapting quickly to a fast-paced consulting environment while applying both **technical** and **strategic problem-solving** to complex security challenges
- Automated processing & validation of vulnerability datasets to improve **PowerBI** reporting used for leadership decision-making
- Partnered with client leadership and cross-functional technology teams to assess, prioritize, and resolve **high-impact vulnerabilities**, ensuring alignment with **enterprise risk management** and compliance requirements

Veermata Jijabai Technological Institute (VJTI) COE CNDS *(Cybersecurity Researcher)*, India *Jun 2023 – Jul 2024*

- Configured **OT & IT systems** from scratch in a research environment using **ESXi** to deploy **SIEM**-capable Security Operations by setting up **ELK** with **IDS – Snort, Suricata, Wazuh,** & **Zeek** for comparative analysis. Implemented security controls on **Cisco L3 switches**, **FortiGate** & **Sophos** firewalls to conduct red-teaming exercises using **MITRE ICS,** working on **Threat Modeling** & **CTI**
- Led Android application security research for **reverse engineering**, utilizing **Binary Ninja, GDB, IDA Pro, and Ghidra** to analyze APK files and **ARM assembly code**, identifying vulnerabilities through **decompiled code & runtime analysis** to secure applications

Cyberpeace Foundation, SAKEC (*Cybersecurity Researcher*), India *Oct 2022 – May 2023*

- Led research initiatives by identifying gaps, devising advanced cybersecurity **strategies**, and presenting **findings** to support the NGO's mission of enhancing **cybersecurity awareness and education** across vulnerable sectors such as **healthcare** & **finance**
- Developed & deployed a **Threat Intelligence Platform** using **open-source** honeypot technologies (Snare, Tanner, Dionaea) to simulate **vulnerabilities**, monitor attacker **behaviors**, & gather actionable **threat data** using security controls in the **educational** sector

PROJECTS

HoneyTrack – Lead for Engineering Project (auti.dev/honeytrack) – Mumbai University, *2021-2024*

- Devised a Honeypot cum **SIEM** tool operating **Python & BASH** scripts combined using **Docker** to automate deployment. Configured **ELK** for log visualization, handling over **20 million logs** from 60 days of **cloud attack** monitoring & improving analysis capabilities
- **Copyrighted** (2022) & published at **IEEE** SCEECS 2023 & Scopus **Journal** of Emerging Technologies & Innovative Research 2023

MobSecOps – An Android Security Framework (auti.dev/mobsecops)– IIT Delhi, *2024*

- Enhanced the Mobile Security Framework (MobSF) by integrating updated modules for static analysis aligned with **OWASP Mobile Top 10 (2023)** vulnerabilities, ensuring comprehensive assessments of **app components, permissions,** and **native libraries**
- Developed advanced **dynamic** analysis features, incorporating **FRIDA scripts** for runtime monitoring, **SSL Certificate Pinning** detection, **root/jailbreak** detection, and **code tampering analysis** to evaluate Android application behavior in real-time
- Integrated **Llama 3 generative AI** model using **HuggingFace,** getting insights by analyzing APK files dynamically, proactively generating patterns, and adapting to evolving attack vectors for enhanced defense against emerging mobile threats

Asset Management for Operational Technology with SIEM & SOAR (auti.dev/elastic) – VJTI, *2023-2024*

- Designed and developed an asset management tool for **Industrial Control System** environments, enabling identification, classification, and real-time monitoring of critical assets while adhering to **IEC 61850 standard**s for **MMS** & **GOOSE** protocols
- Integrated advanced features and functionalities such as **CVSS**-based vulnerability lookup, **endpoint** service identification, device **health** & **threat** monitoring, and **live & PCAP**-based analysis to improve **SOC capabilities** and decision-making
- Enhanced data processing efficiency and **scalability** by implementing **chunking** and **threading** techniques, ensuring quick handling and analysis of capture files over 100 GB in complex industrial applications with an IT data center environment

SKILLS & CERTIFICATIONS

- **Skills:** Penetration Testing, SCADA & Cyber-Physical (Industrial Control) Systems, Cyber Threat Intelligence, Analytics, Risk & Compliance, Network Traffic & Log Analysis, Network Design, Cloud Security, SOC Analysis, Presentation, Consulting
- **Languages & Tools:** Python, Java, C, C++, PowerShell, BASH, Git, VMware ESXi, Proxmox, Windows AD, Terraform, Docker, Kubernetes, Azure, GCP, AWS, MySQL, PostgreSQL, FastAPI, Elasticsearch, Splunk, EZTools, EnCase, KAPE, Autopsy, Volatility, Metasploit, Tenable, Tanium, Microsoft Word, Excel, PowerPoint, PowerBI
- **Frameworks:** ISO/IEC 27001, SOC 2, HIPAA, PCI DSS, CCPA, GDPR, MITRE, Lockheed Martin CKC, CTF (CIA), CIS, CSA, CCM
- **Certifications:** ISC2 CC, EY, Cisco – Networking Essentials, Cloud Security Alliance, Zscaler Workshops, Palo Alto – Cloud, Network Security, & SOC, IBM Cybersecurity & Blockchain, Basel OSINT, Credits for Cryptography & CISSP by Charles Sturt University

ACHIEVEMENTS & LEADERSHIP

- Secured **17th** (4800+ teams) with the USC CTF Team & **123rd** (8000+ individuals) in **National Cyber League** *Fall 2024* across the U.S.
- **1st position** at IIT Delhi's Cybersecurity Hackathon *2024* (Cython) against 500+ teams – **Android Security Framework with GenAI**
- **Runner-up** at AI LA Cerebral Beach Hackathon *2024* Cybersecurity – **Blockchain Smart Contract Auditing using Multiagent AI**
- Presented a **talk** at the **Elasticsearch Community Event** on **Leveraging Cybersecurity using ELK & developing a SIEM tool**